Class 6:

1. Find the Fourier cosine transform of $f(x) = 2t \quad for\, 0 < t < 4$.

   Ans: $\frac{2}{\omega}(1+\cos 4\omega + 2\sin 4\omega)$

2. Find the Fourier cosine transform of $f(x) = \begin{cases} 4t & 0 < t < 1 \\ 4-t & 1 < t < 4 \\ 0 & t > 4 \end{cases}$

   Ans: $\frac{5\cos\omega - 4 - \cos 4\omega}{\omega^2} + \frac{\sin\omega}{\omega}$

3. Find the Fourier cosine transform of $f(t) = te^{-at}, a > 0$ Ans: $\frac{a^2-\omega^2}{(a^2+\omega^2)^2}$

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